Pre-work · The Respiratory System

The Respiratory System. Read the Part A chapter first. The two groupings of the parts, upper and lower, conducting and respiratory, do not line up with each other, and that mismatch is what most questions are built on.

Module 3 · 5 of 5 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the respiratory chapter into mini visuals. Six chunks. One chain carries the airway and one grid carries the mismatch.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A child inhales a small bead. Predict which lung it lodges in and explain the prediction from structure.

Answer. The right lung.

Reasoning. The trachea divides at the carina into two main bronchi, and the two are not mirror images. The right main bronchus is wider and runs more vertically, closer to the line the trachea was already travelling; the left has to angle across to reach around the heart. An object falling with gravity keeps going straight, so it takes the straighter, wider pipe. Notice this is a purely structural answer, and it is also why aspiration pneumonia turns up on the right more often than the left.

Set A · Name it

1. The scroll-shaped projections that swirl air in the nasal cavity are the ______________________.

takes longer than three minutes you are copying instead of organizing.

1 GRID

Two ways of grouping the parts

Structure down the side, from nose to alveolus. Upper or lower in one column, conducting or respiratory in the other. Then circle every row where the two columns disagree. Those rows are the exam questions.

2 HUB

The nasal cavity

One sagittal nasal cavity at the hub. Spoke to the three conchae and their meatuses, the septum with its three parts, the vestibule, the hard and soft palate, the choanae, and the four paranasal sinuses.

3 LADDER

The three regions of the pharynx

Three rungs, nasopharynx at the top. Beside each write what it carries, its epithelium, and its tonsil. The epithelium changes partway down, so mark exactly where.

4 HUB

The larynx

The larynx at the hub. Spoke to all nine cartilages, grouped single and paired, then to the vocal folds, vestibular folds, and glottis. Mark which cartilage is elastic, because only one is.

5 CHAIN

The airway, trachea to alveolus

One chain with arrows, trachea through terminal bronchiole and on into the respiratory zone. Below the chain draw two bars: cartilage decreasing, smooth muscle increasing. Mark where the conducting zone ends.

DRAWING 1

The upper respiratory tract, in sagittal section

BRAIN DUMP FIRST

Two minutes. Everything air passes through before the larynx.

Draw a head cut down the midline. Draw the nasal cavity with all three conchae and the meatus under each, the septum, the vestibule, the hard and soft palate, and the choanae. Mark the frontal and sphenoid sinuses. Then divide the pharynx into its three regions with lines, label each, and mark the epiglottis at the bottom.

In your notebook, head the page **M3-1 The upper respiratory tract, in sagittal section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can trace a breath from the nostril to the larynx on your own drawing, naming every space it crosses.

2. The only complete cartilage ring in the airway is the ____________.
3. The cells that secrete surfactant are the ____________.
4. The last airway of the conducting zone is the ____________.
5. Name the three layers of the respiratory membrane, in the order a gas crosses them.

____

Set B · Predict it

1. The chest wall is punctured. Predict what happens to the lung and name the condition.

____

2. Emphysema destroys alveolar walls. Predict the effect on the exchange surface and say why it is permanent.

____

3. A premature infant lacks surfactant. Predict what happens to the alveoli between breaths.

____

4. Cold air slows the cilia of the respiratory mucosa. Predict the everyday consequence.

____

Set C · Tell it apart

1. The larynx is in the lower respiratory tract but the conducting zone. Explain how both are true.

____

2. Explain why asthma narrows the bronchioles and not the bronchi, using the wall of each.

____

3. Distinguish the true vocal folds from the vestibular folds by position and by job.

6 SPLIT

Right lung against left lung

One split. Lobes, fissures, size, and the feature unique to each. Then hang the pleurae off both branches.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The bronchial tree and the lungs

BRAIN DUMP FIRST

Two minutes. Every branching level, and both lungs with their lobes.

Draw the trachea with its C-shaped cartilage rings and the carina at the bottom. Branch into both main bronchi, and make the right one visibly wider and more vertical. Continue into lobar and segmental bronchi. Now outline both lungs around the tree: three lobes and two fissures on the right, two lobes, one fissure, a cardiac notch, and the lingula on the left.

In your notebook, head the page **M3–2 The bronchial tree and the lungs** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which lung an inhaled object lands in and point at the reason on your own drawing.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

An alveolus and its capillary

BRAIN DUMP FIRST

One minute. Every cell of the alveolar wall and the barrier gases cross.

Draw one alveolar sac with several alveoli opening into it, at large scale. Line the wall with flat type I cells and add a few cuboidal type II cells. Add an alveolar macrophage on the surface. Wrap a capillary against the outside and draw the respiratory membrane where the two meet, labelling all three layers of it.

In your notebook, head the page **M3–3 An alveolus and its capillary** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain, from your drawing, exactly what emphysema destroys and why that reduces gas exchange.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Two ways of grouping the parts

GRID

2 The nasal cavity

HUB

3 The three regions of the pharynx

LADDER

4 The larynx

HUB

5 The airway, trachea to alveolus

CHAIN

6 Right lung against left lung

SPLIT